

SVKM's NMIMS
Mukesh Patel School of Technology Management & Engineering
(Hyderabad Campus)
Computer Engineering Department of Data Science (BTech Sem IV)
Lab Manual
Data Handling and Visualizations

PART A

(PART A: TO BE REFERRED BY STUDENTS)

EXPERIMENT NO.9

A.1 Aim:

To build interactive dashboards in Power BI.

A.2 Outcome:

- Create dashboards
- Add interactivity
- Use slicers and filters effectively

A.3 Datasets Used:

- Campus_Life_Activity_Data_100
- Department_Info

A.4 Concepts Covered:

- Dashboard layout
- Visual interactions
- Drill-down and drill-through
- Slicers

A.5 Tasks:

1. Create multiple visuals
2. Add slicers and filters
3. Enable drill-down functionality
4. Design a dashboard layout
5. Ensure interactivity between visuals

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PART B

Student Details:

- **Roll No.:** A046
- **Name:** A. Sahithi
- **Class:** CSDS-A
- **Batch:** A2
- **Date of Experiment:**
- **Date of Submission:**
- **Grade:**

B.1 Documentation Written by Student:

The screenshot displays a data visualization tool interface. On the left, a sidebar shows 'Queries [2]' with 'Campus_Life_Activity_...' and 'Department_Info 2'. The main area features a table with 23 rows and 10 columns. The columns are: 'Date', 'Day', 'Department', 'Activity_Type', 'Attendance_Count', 'Avg_Mood_Score', 'WiFi_Usage...', 'Cafeteria_S...', and 'Time_S...'. The table data includes dates from 4/1/2025 to 18/1/2025, departments like CSE, MBA, DS, AI, ECE, and various activities such as Event, Library, Lab, Class, Sports, and Morning/Evening/Afternoon sessions. On the right, the 'Query settings' panel is visible, showing 'Properties' for 'Campus_Life_Activity_...' and 'Applied steps' including 'Source', 'Promoted ...', and 'Changed c...'. The table formula bar at the top reads: 'Table.TransformColumnTypes("#Promoted headers", {{"Date", type date}, {"Day", type text}, {"Department", type text}, ...)'.

	Date	Day	Department	Activity_Type	Attendance_Count	Avg_Mood_Score	WiFi_Usage...	Cafeteria_S...	Time_S...
1	4/1/2025	Saturday	CSE	Event	145	2.7	60.24	6043	Evening
2	18/1/2025	Saturday	CSE	Library	236	2.1	12.03	2405	Evening
3	20/1/2025	Monday	CSE	Library	121	4.1	57.6	3936	Morning
4	15/1/2025	Wednesd...	MBA	Event	23	4.3	16.97	3962	Afternoon
5	9/1/2025	Thursday	AI	Lab	192	2.3	33.49	3440	Afternoon
6	20/1/2025	Monday	DS	Class	255	3.6	77.98	3600	Morning
7	18/1/2025	Saturday	DS	Library	205	3.7	67.84	875	Evening
8	8/1/2025	Wednesd...	DS	Class	139	4.6	33.51	4214	Evening
9	12/1/2025	Sunday	AI	Event	201	2.6	25.02	6099	Evening
10	3/1/2025	Friday	MBA	Lab	293	4.2	17.26	3608	Afternoon
11	18/1/2025	Saturday	AI	Event	48	2.7	7.41	3084	Afternoon
12	9/1/2025	Thursday	CSE	Lab	181	2.6	42.44	7747	Evening
13	15/1/2025	Wednesd...	AI	Event	91	2.7	47.1	2652	Evening
14	19/1/2025	Sunday	ECE	Library	224	3.1	79.8	1633	Evening
15	16/1/2025	Thursday	CSE	Class	76	2.5	17	6074	Afternoon
16	20/1/2025	Monday	CSE	Sports	215	3.8	40.1	2559	Evening
17	1/1/2025	Wednesd...	CSE	Library	156	4.3	30.51	2904	Afternoon
18	6/1/2025	Monday	ECE	Class	154	4.9	62.15	4658	Morning
19	10/1/2025	Friday	MBA	Library	121	2.5	62.19	4918	Evening
20	1/1/2025	Wednesd...	MBA	Event	270	2.1	74.68	7698	Afternoon
21	8/1/2025	Wednesd...	CSE	Lab	60	2.3	41.45	1066	Evening
22	5/1/2025	Sunday	AI	Sports	104	2.8	70.43	3966	Morning
23	18/1/2025	Saturday	AI	Event	224	5	53.74	4088	Evening

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Power Query [2]

Table.TransformColumnTypes(#"Promoted headers", {{{"Department", type text}, {"HOD_Name", type text}, {"Total_Students", type number}, {"Budget_Allocated", type number}}})

	Department	HOD_Name	Total_Students	Budget_Allocated
1	CSE	Dr. Sharma	650	1500000
2	IT	Dr. Verma	500	1100000
3	ECE	Dr. Rao	550	1300000
4	ME	Dr. Patel	450	900000

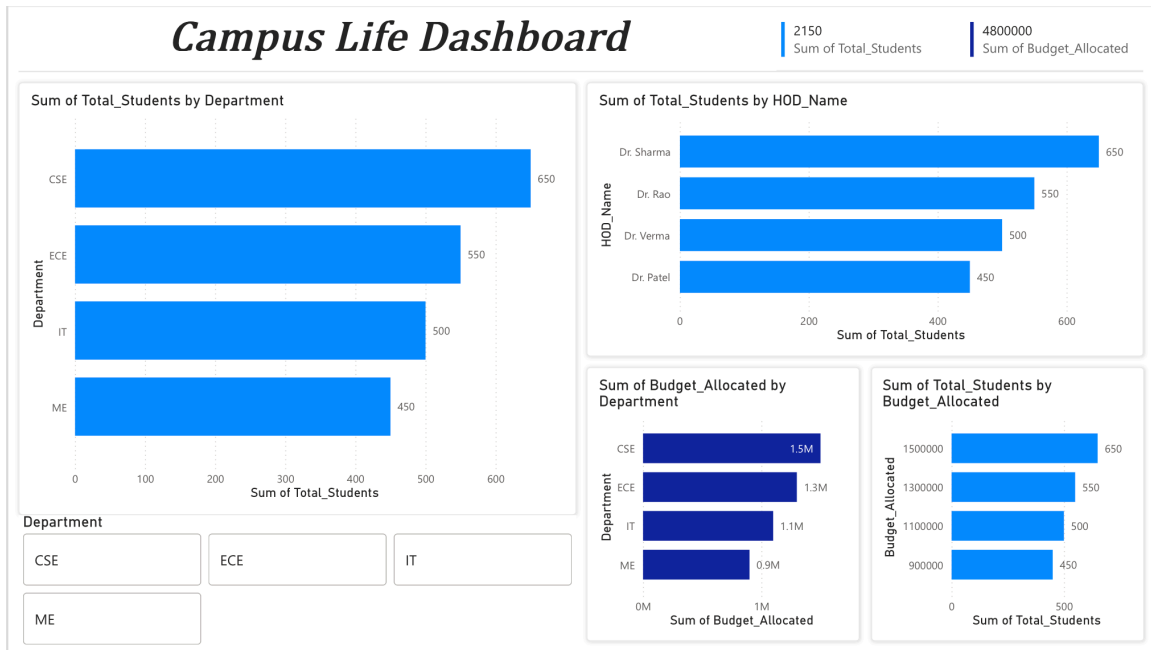
Query settings

Properties

Name
Department_Info 2

Applied steps

Source
Promoted ...
Changed c...



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B.2 Observations and Learning:

- Learned how to create interactive dashboards in Power BI
- Understood the importance of slicers and filters in data analysis
- Observed how different visuals provide better insights when combined
- Gained knowledge about drill-down and drill-through features
- Improved skills in dashboard design and layout structuring

B.3 Conclusion:

This experiment helped in understanding how to build interactive dashboards using Power BI. By using slicers, filters, and drill features, meaningful insights were derived from the dataset. The experiment enhanced visualization skills and demonstrated the importance of interactivity in business intelligence tools.